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Factoring

Algebra Review - Part 5

Required Video: Check on website

Pre-requisite Videos: Factoring (All)

NOTE: If you have a leading negative; factor out the negative first.

$$(1.) \quad -13k^2 - 52 \quad \text{Factor by GCF} \quad \text{GCF} = -13 \\ -13(k^2 + 4)$$

$$(2.) \quad -6a^5 + 12a^4 - 9a^3 \quad \text{Factor by GCF} \quad \text{GCF} = -3a^3 \\ -3a^3(2a^2 - 4a + 3)$$

$$(3.) \quad 9p^2 + 27p^3 - 18p^4 \quad \text{Factor by GCF} \\ \text{Arrange in descending order of exponents} \\ -18p^4 + 27p^3 + 9p^2 \\ -9p^2(2p^2 - 3p - 1) \quad \text{GCF} = -9p^2$$

$$(4.) \quad (k-5)2k + (k-5)3 \quad \text{Factor by GCF; GCF} = (k-5) \\ 2k(k-5) + 3(k-5) \\ (k-5)[2k+3] \\ (k-5)(2k+3)$$

$$(5.) \quad 3c(c^2+5) + 7(c^2+5) \quad \text{Factor by GCF; GCF} = c^2+5 \\ (c^2+5)[3c+7] = (c^2+5)(3c+7)$$

(6.) $6g(g-3) - 2(g-3)$ Factor by GCF; GCF = $(g-3)$
 $(g-3)[6g-2]$
 $(g-3)(6g-2)$

(7.) $6a^3 - 9a^2 - 2a + 3$ Factor by Grouping
 $6a^3 - 9a^2 = 3a^2(2a-3)$ I prefer the Vertical Method
 $-2a + 3 = -1(2a-3)$ but you can use the Horizontal
 $(2a-3)(3a^2-1)$ Method if you wish.

(8.) $3y^2 - xy - 6y + 2x$ Factor by Grouping
 $3y^2 - xy = y(3y-x)$
 $-6y + 2x = -2(3y-x)$
 $(3y-x)(y-2)$

(9.) $4d^3 + 6d^2 - 2d - 3$ Factor by Grouping
 $4d^3 + 6d^2 = 2d^2(2d+3)$
 $-2d - 3 = -1(2d+3)$
 $(2d+3)(2d^2-1)$

(10.) $-k^2 + 13k - 42$ Factor Quadratic Trinomials
 $-1(k^2 - 13k + 42)$
 $k^2 - 13k + 42$ $(k^2)(42) = 42k^2$
Factors are $-6k$ and $-7k$
 $k^2 - 6k - 7k + 42$
 $k^2 - 6k = k(k-6)$
 $-7k + 42 = -7(k-6)$

(10.) $(k-6)(k-7)$
 $-1(k-6)(k-7)$

(11.) $q^2 + 12 + 13q$ Factor Quadratic Trinomials
 $q^2 + 13q + 12$ $(q^2)(12) = 12q^2$
 Factors are q and $12q$
 $q^2 + q + 12q + 12$
 $q^2 + q = q(q+1)$
 $12q + 12 = 12(q+1)$
 $(q+1)(q+12)$

(12.) $m^2 + 10m + 15$ Factor Quadratic Trinomials
 $(m^2)(15) = 15m^2$ Factors exist but roots are irrational
 because the discriminant is not a
 Perfect Square.

Therefore, the trinomial cannot be solved by factoring.
 Trinomial is prime.

General form of a Quadratic Trinomial is $ax^2 + bx + c$
 Discriminant = $b^2 - 4ac$

(a.) If the ~~Discriminant~~ is positive (which means $b^2 - 4ac > 0$)
 then the Quadratic Equation has 2 real roots.

If the Discriminant is positive, ^{but} ~~and~~ is a perfect square;
 then the Quadratic Trinomial can be solved by Factoring
 and the Quadratic Equation has 2 rational roots.

- (a) If the Discriminant is positive, but is not a perfect square; then the Quadratic Trinomial cannot be solved by Factoring and the Quadratic Equation has 2 irrational roots.
- (b) If the Discriminant is zero ($b^2 - 4ac = 0$) then the Quadratic Trinomial has one repeated factor and the Quadratic Equation has one repeated real root.
- (c) If the Discriminant is negative ($b^2 - 4ac < 0$) then the Quadratic Trinomial cannot be solved by Factoring and the Quadratic Equation has 2 complex roots.

(13.) $6K^2 + K - 35$ Factor Quadratic Trinomials
 $(6K^2)(-35) = -210K^2$ Factors are $15K$ and $-14K$
 $6K^2 + 15K - 14K - 35$
 $6K^2 + 15K = 3K(2K + 5)$
 $-14K - 35 = -7(2K + 5)$
 $(2K + 5)(3K - 7)$

Check By FOIL or Vertical Method or Box Method or Horizontal Method

F:	$(2K)(3K) = 6K^2$	
O:	$(2K)(-7) = -14K$	$6K^2 - 14K + 15K - 35$
I:	$(5)(3K) = 15K$	$6K^2 + K - 35$
L:	$(5)(-7) = -35$	

(14.) $15x^2 - 22x - 48$ Factor Quadratic Trinomials
 $(15x^2)(-48) = -720x^2$ Factors are $-40x$ and $18x$
 $15x^2 - 40x + 18x - 48$
 $15x^2 - 40x = 5x(3x - 8)$
 $18x - 48 = 6(3x - 8)$
 $(3x - 8)(5x + 6)$

(15.) $20y^2 + 53y + 18$ Factor Quadratic Trinomials
 $(20y^2)(18) = 360y^2$ Factors are $8y$ and $45y$
 $20y^2 + 8y + 45y + 18$
 $20y^2 + 8y = 4y(5y + 2)$
 $45y + 18 = 9(5y + 2)$
 $(5y + 2)(4y + 9)$

(16.) $p^2 - 18p + 81$ Factor Quadratic Trinomials
 $(p^2)(81) = 81p^2$ Factors are $-9p$ and $-9p$
 $p^2 - 9p - 9p + 81$
 $p^2 - 9p = p(p - 9)$
 $-9p + 81 = -9(p - 9)$
 $(p - 9)(p - 9) = (p - 9)^2$

(17.) $16m^2 + 40m + 25$ Factor Quadratic Trinomials
 $(16m^2)(25) = 400m^2$ Factors are $20m$ and $20m$
 $16m^2 + 20m + 20m + 25$
 $16m^2 + 20m = 4m(4m + 5)$
 $20m + 25 = 5(4m + 5)$
 $(4m + 5)(4m + 5) = (4m + 5)^2$

(18.) $25h^2 - 60h + 36$ Factor Quadratic Trinomials
 $(5h^2)(36) = 90h^2$ Factors are $-30h$ and $-30h$
 $25h^2 - 30h - 30h + 36$
 $25h^2 - 30h = 5h(5h - 6)$
 $-30h + 36 = -6(5h - 6)$
 $(5h - 6)(5h - 6) = (5h - 6)^2$

(19.) $y^2 + 12y + 36$ Factor Quadratic Trinomials
 $(y^2)(36) = 36y^2$ Factors are $6y$ and $6y$
 $y^2 + 6y + 6y + 36$
 $y^2 + 6y = y(y + 6)$
 $6y + 36 = 6(y + 6)$
 $(y + 6)(y + 6) = (y + 6)^2$

~~(20.)~~ Difference of Two Squares
 $x^2 - y^2 = (x + y)(x - y)$ and vice versa

Difference of Two Cubes
 $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

Sum of Two Cubes
 $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

(20.) $4k^2 - 25$ Factor by Diff. of Two Sq.
 $2^2k^2 - 5^2$
 $(2k)^2 - 5^2$ Compare to $x^2 - y^2$; $x = 2k$; $y = 5$
 $(2k+5)(2k-5)$

(21.) $x^2 - 64$ Factor by Diff. of Two Sq.
 $x^2 - 8^2$ Compare. $x = x$, $y = 8$
 $(x+8)(x-8)$

(22.) $50p^2 - 72$ GCF; then Diff. of Two Sq.
 $2(25p^2 - 36)$
 $2(5^2p^2 - 6^2)$
 $2[(5p)^2 - 6^2]$ Compare. $x = 5p$, $y = 6$
 $2[(5p+6)(5p-6)]$
 $2(5p+6)(5p-6)$

(23.) $m^2 - 17$ Diff. of Two Sq.
 We know 17 is not a perfect square. But, we can put it
 as a square of $\sqrt{17}$
 $m^2 - (\sqrt{17})^2$ Compare. $x = m$, $y = \sqrt{17}$
 $(m+\sqrt{17})(m-\sqrt{17})$

(24.) $25v^2 - \frac{1}{49}$ Diff. of Two Sq.
 $5^2v^2 - \frac{1}{7^2}$

$$(24.) \quad 52v^2 - \frac{12}{7^2}$$

$$(5v)^2 - \left(\frac{1}{7}\right)^2 \quad \text{Compare. } x=5v, y=\frac{1}{7}$$

$$\left(5v + \frac{1}{7}\right)\left(5v - \frac{1}{7}\right)$$

$$(25.) \quad e^4 - 1$$

Diff. of Two Sq.

$$(e^2)^2 - 1^2$$

Compare. $x=e^2, y=1$

$$(e^2+1)(e^2-1)$$

For the e^2-1

Diff. of Two Sq. again

$$e^2 - 1^2$$

Compare. $x=e, y=1$

$$(e+1)(e-1)$$

$$\therefore e^4 - 1 = (e^2+1)(e+1)(e-1)$$

$$(26.) \quad 16x^4 - 81$$

Diff. of Two Sq.

$$4^2(x^2)^2 - 9^2$$

$$(4x^2)^2 - 9^2$$

Compare. $x=4x^2, y=9$

$$(4x^2+9)(4x^2-9)$$

For the $4x^2-9$

Diff. of Two Sq. again

$$2x^2 - 3^2$$

$$(2x)^2 - 3^2$$

Compare. $x=2x, y=3$

$$(2x+3)(2x-3)$$

$$\therefore 16x^4 - 81 = (4x^2+9)(2x+3)(2x-3)$$

(27.)

$$x^3 - 8$$

$$x^3 - 2^3$$

$$(x-2)(x^2 + 2x + 2^2)$$

$$(x-2)(x^2 + 2x + 4)$$

Diff. of Two Cubes

Compare. $x=x, y=2$

(28.)

$$x^3 + 64$$

$$x^3 + 4^3$$

$$(x+4)(x^2 - 4x + 4^2)$$

$$(x+4)(x^2 - 4x + 16)$$

Sum of Two Cubes

Compare. $x=x, y=4$

(29.)

$$27p^3 - 125$$

$$3^3p^3 - 5^3$$

$$\cancel{27p^3} (3p)^3 - 5^3$$

$$(3p-5)((3p)^2 + (3p)(5) + 5^2)$$

$$(3p-5)(9p^2 + 15p + 25)$$

Diff. of Two Cubes

Compare. $x=3p, y=5$

(30.)

$$k^3 + \frac{8}{27}$$

$$k^3 + \frac{2^3}{3^3}$$

$$k^3 + \left(\frac{2}{3}\right)^3$$

Sum of Two Cubes

Compare. $x=k, y=\frac{2}{3}$

$$\left(k + \frac{2}{3}\right)\left(k^2 - (k)\left(\frac{2}{3}\right) + \left(\frac{2}{3}\right)^2\right)$$

$$\left(k + \frac{2}{3}\right)\left(k^2 - \frac{2k}{3} + \frac{4}{9}\right)$$

(31.) $h^3 - 0.343$

$h^3 - (0.7)^3$

$(h-0.7)(h^2 + 0.7h + (0.7)^2)$

$(h-0.7)(h^2 + 0.7h + 0.49)$

Diff. of Two Cubes

Compare. $x=h, y=0.7$

(32.) $-2y^4 + 54y$

$-2y(y^3 + 27)$

For $y^3 + 27$

$y^3 + 3^3$

$(y+3)(y^2 - 3y + 3^2)$

$(y+3)(y^2 - 3y + 9)$

$\therefore -2y^4 + 54y = -2y(y+3)(y^2 - 3y + 9)$

GCF; and Sum of Two Cubes

Compare. $x=y, y=3$

(33.) $-2y^4 + 54y$

$-2y(y^3 - 27)$

For $y^3 - 27$

$y^3 - 3^3$

$(y-3)(y^2 + 3y + 9)$

$\therefore -2y^4 + 54y = -2y(y-3)(y^2 + 3y + 9)$

GCF and Diff. of Two Cubes

Compare. $x=y, y=3$

(34.) $25 - 16c^2$

$5^2 - 4^2c^2$

$5^2 - (4c)^2$

$(5+4c)(5-4c)$

Diff. of Two Sq.

Compare. $x=5, y=4c$

(35.)

$$x^4 - 10x^2 + 9$$

Factor Quartic Trinomials

$$(x^4)(9) = 9x^4$$

Factors are $-x^2$ and $-9x^2$

$$x^4 - x^2 - 9x^2 + 9$$

$$x^4 - x^2 = x^2(x^2 - 1)$$

$$-9x^2 + 9 = -9(x^2 - 1)$$

$$(x^2 - 1)(x^2 - 9)$$

$$(x^2 - 1^2)(x^2 - 3^2)$$

$$(x+1)(x-1)(x+3)(x-3)$$

(36.)

$$p^4 + 13p^2 + 36$$

Factor Quartic Trinomials

$$(p^4)(36) = 36p^4$$

Factors are $9p^2$ and $4p^2$

$$p^4 + 9p^2 + 4p^2 + 36$$

$$p^4 + 9p^2 = p^2(p^2 + 9)$$

$$4p^2 + 36 = 4(p^2 + 9)$$

$$(p^2 + 9)(p^2 + 4)$$

(37.)

$$k^6 - 7k^3 - 8$$

Factor Hexic or Sextic Trinomials

$$(k^6)(-8) = -8k^6$$

Factors are k^3 and $-8k^3$

$$k^6 + k^3 - 8k^3 - 8$$

$$k^6 + k^3 = k^3(k^3 + 1)$$

$$-8k^3 - 8 = -8(k^3 + 1)$$

$$(k^3 + 1)(k^3 - 8)$$

$$(k^3 + 1^3)(k^3 - 2^3)$$

$$(k+1)(k^2 - k + 1)(k-2)(k^2 + 2k + 4)$$

$$(k+1)(k-2)(k^2 - k + 1)(k^2 + 2k + 4)$$

(38.) $2(g+3)^2 + 3(g+3) - 54$ Factor Quadratic Trinomials
Let $p = g+3$.

$$2p^2 + 3p - 54$$

$$(2p^2)(-54) = -108p^2 \text{ Factors are } 12p \text{ and } -9p$$

$$2p^2 + 12p - 9p - 54$$

$$2p^2 + 12p = 2p(p+6)$$

$$-9p - 54 = -9(p+6)$$

$$(p+6)(2p-9)$$

But, recall that $p = g+3$

$$((g+3)+6)(2(g+3)-9)$$

$$(g+3+6)(2g+6-9)$$

$$(g+9)(2g-3)$$

(39.) $y^6 - 26y^3 - 27$ Factor Hexic Trinomials

$$(y^6)(-27) = -27y^6 \text{ Factors are } y^3 \text{ and } -27y^3$$

$$y^6 + y^3 - 27y^3 - 27$$

$$y^6 + y^3 = y^3(y^3 + 1)$$

$$-27y^3 - 27 = -27(y^3 + 1)$$

$$(y^3 + 1)(y^3 - 27)$$

$$(y^3 + 1^3)(y^3 - 3^3)$$

$$(y+1)(y^2 - y + 1)(y-3)(y^2 + 3y + 3^2)$$

$$(y+1)(y^2 - y + 1)(y-3)(y^2 + 3y + 9)$$

$$(y+1)(y-3)(y^2 - y + 1)(y^2 + 3y + 9)$$

(40.) $3(x+5)^2 + 2(x+5) - 21$ Factor Quadratic Trinomials

$$\text{Let } q = x+5$$

$$3q^2 + 2q - 21$$

$$(3q^2 - 21) = -63q^2 \quad \text{Factors are } -7q \text{ and } 9q$$

$$3q^2 - 7q + 9q - 21$$

$$3q^2 - 7q = q(3q - 7)$$

$$9q - 21 = 3(3q - 7)$$

$$(3q - 7)(q + 3)$$

But, recall that $q = x+5$

$$(3(x+5) - 7)((x+5) + 3)$$

$$(3x + 15 - 7)(x + 5 + 3)$$

$$(3x + 8)(x + 8)$$

(41.) $-30K - 4K^2 + 2K^3$ GCF; and Quadratic Trinomials

$$2K^3 - 4K^2 - 30K \quad \text{GCF} = 2K$$

$$2K(K^2 - 2K - 15)$$

$$\text{For } K^2 - 2K - 15$$

$$(K^2 - 15) = -15K^2 \quad \text{Factors are } 3K \text{ and } -5K$$

$$K^2 + 3K - 5K - 15$$

$$K^2 + 3K = K(K + 3)$$

$$-5K - 15 = -5(K + 3)$$

$$(K + 3)(K - 5)$$

$$\therefore -30K - 4K^2 + 2K^3 = 2K(K + 3)(K - 5)$$

(42.) $-28y^3 + 16y^2 + 80y$ GCF; and Quadratic Trinomials

$$-4y(7y^2 - 4y - 20)$$

$$\text{For } 7y^2 - 4y - 20$$

(42.) $(7y^2)(-20) = -140y^2$ Factors are $-14y$ and $10y$

$$7y^2 - 14y + 10y - 20$$

$$7y^2 - 14y = 7y(y - 2)$$

$$10y - 20 = 10(y - 2)$$

$$(y - 2)(7y + 10)$$

$$\therefore -28y^3 + 16y^2 + 80y = -4y(y - 2)(7y + 10)$$

(43.) $r^3 + 3r^2 - 4r - 12$ Grouping, and Diff. of Two Sq.

$$r^3 + 3r^2 = r^2(r + 3)$$

$$-4r - 12 = -4(r + 3)$$

$$(r + 3)(r^2 - 4)$$

$$(r + 3)(r^2 - 2^2)$$

$$(r + 3)(r + 2)(r - 2)$$

References.

Coburn, J., and Coffelt, J. (2014). College algebra essentials (Third edition).